

The Isotropic 3-Dimensional Quantum Oscillator as a Molecular Model

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Abstract

The solution for the three dimensional quantum oscillator is reviewed here and then the results used as a molecular model to study the group 6 hexafluorides (XF_6). We show how that configurations for this molecule are consistent with the $3p$ state of the oscillator. Known bond lengths for the hexafluoride molecule are compared with the results from the radial equation and are shown to be consistent for a particular oscillation frequency, which is affiliated with very low temperatures. It is hoped that future work will lead to this model being used to easily arrive at results that formally would require dealing with the cumbersome many electron, multi-nuclear Schrödinger equation.

Keywords: Isotropic oscillator; Hexa-fluoride; Laguerre's equation; Radial function; Molecular model; Low temperature model

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1 Introduction

The three dimensional (3D) quantum harmonic oscillator has been used in the past to model a variety of low energy systems. These include the study of elementary particles coalescing into light nuclei and hypernuclei [1,2], the combining of hadrons into hadronic molecules [3] and recently, Kordell, Fries and Ko applied this model to the study of the coalescing of two indistinguishable non-relativistic particles represented by wave packets [4]. We show in this report, that the 3D oscillator model provides for a reasonable description for the low temperature configuration of the highly symmetric group 6 hexafluorides molecules.

We begin by reviewing the solution for the three dimensional isotropic quantum oscillator. The result is then applied to the study of the group 6 hexafluorides of O_h symmetry. We show how configurations for this molecule are consistent with the certain angular momentum states of the 3D oscillator. Known bond lengths for this molecule are compared with the results from the radial equation and are shown to be consistent for a particular oscillation frequency, which is affiliated with very low temperatures.

We begin our derivation with the time-independent Schrödinger equation

$$\left(-\frac{\hbar^2}{2\mu}\nabla^2 + V\right)\Psi = E\Psi, \quad (1)$$

where $\hbar$ is Planck's constant, and μ the reduced mass. We assume that the state-functions, Ψ , take the separable form, $\Psi = \Phi(\phi)\Theta(\theta)R(r)$, in spherical coordinates, where, due to the symmetry of the 3D oscillator, the angular solution yields the well-known spherical harmonics, Y_l^m , (i.e. $Y_l^m(\theta, \phi) = \Theta_l(\theta)\Phi_n(\phi)$). We will therefore focus in the next section on the solution to the radial portion of Ψ , $R(r)$.

2 Radial Function Derivation

After operating on Ψ with the Hamiltonian in Eq. (1), and letting $u = rR$, we find that

$$-\frac{\hbar}{2\mu}\frac{d^2u}{dr^2} + \left(\frac{\hbar^2 l(l+1)}{2\mu r^2} + V - E\right) = 0, \quad (2)$$

where l is the angular momentum quantum number. Now the 3D oscillator potential is

$$V = \frac{1}{2}\mu\omega^2 r^2. \quad (3)$$

Using this potential in Eq. (2) we get

$$-\frac{\hbar^2}{2\mu}\frac{d^2u}{dr^2} + \left(\frac{\hbar^2 l(l+1)}{2\mu r^2} + \frac{1}{2}\mu\omega^2 r^2 - E\right)u = 0. \quad (4)$$

Factoring and re-arranging leads to,

$$-\frac{\hbar^2}{2\mu}\left(\frac{d^2u}{dr^2} - \frac{l(l+1)u}{r^2} - \frac{\mu^2\omega^2 r^2 u}{\hbar^2} + \frac{2E\mu}{\hbar^2}u\right) = 0. \quad (5)$$

We make the simplifying substitution

$$\beta = \frac{\omega\mu}{\hbar} . \quad (6)$$

Using β in Eq. (5) and multiplying through by $-2\mu/\hbar^2$ yields

$$\frac{d^2u}{dr^2} - \frac{l(l+1)u}{r^2} - \beta^2 r^2 u + \frac{2\mu E}{\hbar^2} u = 0 . \quad (7)$$

Now, we use the fact that $\mu = \hbar\beta/\omega$, to eliminate the reduced mass in Eq. (7) which leads to

$$\frac{d^2u}{dr^2} - \frac{l(l+1)u}{r^2} - \beta^2 r^2 u + \frac{2\beta E u}{\hbar\omega} = 0 . \quad (8)$$

Finally this result is divided through by β to obtain a convenient form:

$$\frac{d^2u}{\beta dr^2} - \frac{l(l+1)u}{\beta r^2} - \beta r^2 u + \frac{2Eu}{\hbar\omega} = 0 \quad (9)$$

Now, as we are dealing with a 3D harmonic oscillator we expect that $E = \hbar\omega(n + 3/2)$, where $n = 0, 1, 2, 3, \dots$ so that Eq. (9) becomes

$$\frac{d^2u}{\beta dr^2} - \frac{l(l+1)u}{\beta r^2} - \beta r^2 u + 2\left(n + \frac{3}{2}\right)u = 0 . \quad (10)$$

The variable substitution $\rho = \beta r^2$ is now made. This gives

$$\frac{d^2u}{dr^2} = 2\frac{du}{d\rho} + 4\rho\frac{d^2u}{d\rho^2} . \quad (11)$$

Inserting this result into Eq. (10) and dividing through by 4ρ yields

$$\frac{d^2u}{d\rho^2} + \frac{1}{2\rho}\frac{du}{d\rho} - \frac{l(l+1)u}{4\rho^2} - \frac{u}{4} + \frac{(n + \frac{3}{2})u}{2\rho} = 0 . \quad (12)$$

A solution for the above differential equation can be obtained by considering the solution in the limiting cases of $\rho \rightarrow \infty$ and $\rho \rightarrow 0$. First we consider Eq. (12) when $\rho \rightarrow \infty$. The following terms will remain:

$$\frac{d^2u}{d\rho^2} - \frac{u}{4} = 0 . \quad (13)$$

This is just the second order differential equation with constant coefficients. From the two term general solution we retain the term that satisfies our boundary conditions, that is

$$u = e^{-\rho/2} . \quad (14)$$

Now we examine Eq. (12), when $\rho \rightarrow 0$. In this case we retain terms of second order to get:

$$\frac{d^2u}{d\rho^2} - \frac{l(l+1)u}{4\rho^2} + \frac{1}{2\rho}\frac{du}{d\rho} = 0 . \quad (15)$$

Multiplying through by $4\rho^2$ we arrive at a Cauchy-Euler equation:

$$4\rho^2 \frac{d^2 u}{d\rho^2} + 2\rho \frac{du}{d\rho} - l(l+1)u = 0 . \quad (16)$$

Through well-know solution techniques, [5], a two term general solution for Eq. (16) can be obtained. From these, the solution that is consistent with out boundary conditions is $\rho^{(l+1)/2}$.

Therefore, we propose that the complete solution to Eq. (12) is of the form:

$$u = e^{-\rho/2} \rho^{(1+l)/2} v(\rho) , \quad (17)$$

where $v(\rho)$ is some yet to be determined function. Using Eq. (17) to compute $du/d\rho$ and $d^2u/d\rho^2$ and inserting the results back into Eq. (12) after some manipulation one can arrive at

$$\frac{e^{-\rho/2}}{2} \rho^{(l-1)/2} \left(2\rho \frac{d^2 v}{d\rho^2} + (2l+3-2\rho) \frac{dv}{d\rho} + (n-l)v \right) = 0 . \quad (18)$$

Rearranging and dividing by 2 yields:

$$\rho \frac{d^2 v}{d\rho^2} + (l + \frac{3}{2} - \rho) \frac{dv}{d\rho} + \frac{1}{2}(n-l)v = 0 . \quad (19)$$

It can be seen that Eq. (19) is in the form of the *associated Laguerre differential equation* [6]:

$$\left[\rho \frac{d^2}{d\rho^2} + (j+1-\rho) \frac{d}{d\rho} + q \right] L_q^j(\rho) = 0 , \quad (20)$$

defined for $q > 0$ and $j > -1$, where L_q^j are the associated Laguerre polynomials. On comparing Eq. (19) and Eq. (20) one finds that

$$j+1 = l + \frac{3}{2} \rightarrow j = l + \frac{1}{2} , \quad (21)$$

and

$$q + \left(l + \frac{1}{2} \right) = \frac{1}{2}(n-l) \rightarrow q = \frac{(n-l)}{2} , \quad (22)$$

so that the associated Laguerre polynomials are solutions for Eq. (19) when these assignments for j and q are made. Therefore we have that

$$v(\rho) = L_{(n-l)/2}^{l+1/2}(\rho) , \quad (23)$$

where l is limited by the requirement $n-l > 0$.

Now, using Eq. (23), along with the fact that $\rho = \beta r^2$ and $u = rR$, in Eq. (17), we arrive at the final expression for our radial function

$$R(r) = N r^l e^{(\beta r^2)/2} L_{(n-l)/2}^{l+1/2}(\beta r^2) , \quad (24)$$

where N is a normalization constant. The final overall state function is then

$$\Psi_{nlm} = N Y_l^m(\theta, \phi) r^l e^{(-\beta r^2)/2} L_{(n-l)/2}^{l+1/2}(\beta r^2) , \quad (25)$$

where we have used the normalized spherical harmonics.

To investigate the radial locations of peak density it is useful to define the radial distribution function D , as

$$D = 4\pi r^2 R^2 . \quad (26)$$

In the next section we will consider an application of these results as a model for a low energy state of an atomic molecule of high symmetry.

3 Molecular Model

Now we will use the previous result as a molecular model for the group 6 hexafluorides of O_h symmetry. Typical examples are UF_6 , WF_6 , and MoF_6 . This is possible as the angular momentum states of the 3D oscillator are identical to those of the hydrogen atom which then yield probability densities showing distinct regions of coalescence for the mass of the oscillator, many of which lack angular symmetry. A number of these states have densities that are consistent with something that would describe the location of atoms in molecules of well known conformations. We suggest that an excellent example of this correlation is with the group 6 hexafluorides.

Though s states possess the symmetry to contain such a molecule, we find intriguing the Ψ_{320} state, which seems to provide the correct probability density to hold the atoms of the hexafluoride molecules previously mentioned. This is depicted in Figure 1.

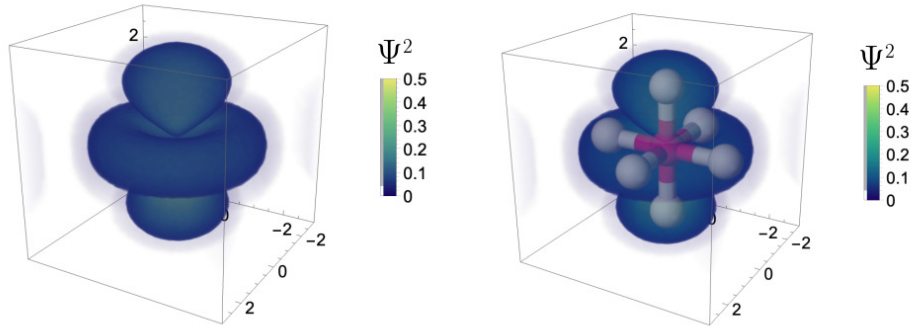


Figure 1: Probability distributions for the Ψ_{320} state of the 3D oscillator. On the right, a ball and stick model of a group 6 hexafluoride is inserted in the distribution region. Distances are in angstroms.

Assuming that the Ψ_{320} state is descriptive of such a molecule we can now use the parameters involved in the derivation of the previous section to establish a physical connection between the model and molecule. We initiate this process by discovering how the parameter values involved relate to the known bond lengths of the molecule. For this process, it is useful to utilize the density distribution function D defined in Eq. (26). We desire the radial location for the maxima of D that is the r for which $dD/dr = 0$. In particular, we select the one at greatest density to be the value for the known bond length.

Now, the constants in β must be defined. We let the reduced mass μ be given by

$$\mu = \frac{6m_X m_F}{m_X + 6m_F} , \quad (27)$$

where m_X is the mass of the group 6 element and m_F the mass of fluorine. Now, we get the known average bond length between atoms in these molecules from a table [7]. The frequency ω is left as a free parameter and adjusted. Values used in the process are listed in Table 1.

ω	$1.50796 \times 10^9 \text{ (rad/s)}$
β	$0.449349 \text{ (\AA)}^{-2}$
μ	$4.9817 \times 10^{-25} \text{ (kg)}$
bond length (W-F)	$1.832 \text{ (\AA)}$
T	11.456 mK

Table 1: Values for the Ψ_{320} state for the radial function.

An example of the D plotted for the case of WF_6 is shown in Figure 2.

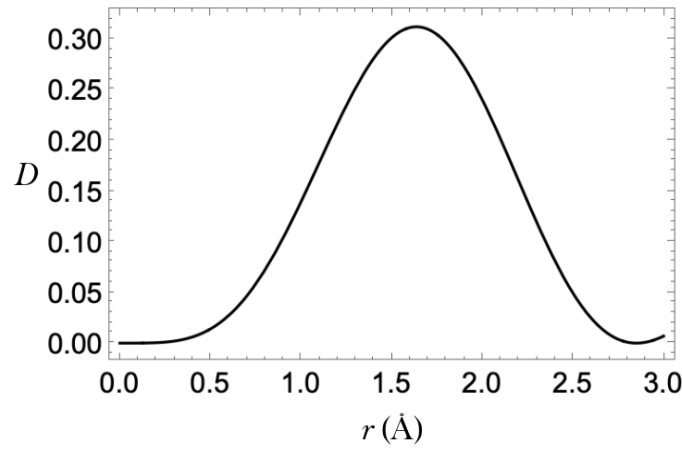


Figure 2: Radial distribution function, D , plotted for W-F. Parameter values used are listed in Table 1.

Interestingly, we a frequency in the neighborhood of 10^9 rad/s , this model is descriptive of a very low energy state for the molecule. For example, letting $k_B T = \hbar \omega$ leads to a temperature of around 11 mK.

4 Conclusion

In this study we reviewed the solution for the radial equation in the 3D quantum harmonic oscillator. The resulting state function was then used as a molecular model for the very

symmetric group 5 hexafluorides. We find that the model is descriptive of the general geometry of such molecules at very low temperatures.

Much work still remains to be done. It is our hope that this low temperature model can be used to analyze the molecule at room temperature. It is well-known that the electronic hamiltonian for such a molecule would contain too many terms to be dealt with analytically and therefore approximation methods are always resorted to. Whereas, this low temperature oscillator model is fully determined in terms of known special functions. Could aspects of the low temperature model be extrapolated classically to the room temperature world and used to estimate quantities of interest for the gaseous molecule at room temperature [8] ?

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